RAJALAKSHMI INSTITUTE OF TECHNOLOGY
(An Autonomous Institution, Affiliated to Anna University, Chennai)

DEPARTMENT OF CSE (ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING)

ACADEMIC YEAR 2025 - 2026

SEMESTER III

DATA ANALYTICS LABORATORY

MINI PROJECT REPORT

REGISTER NUMBER	2117240030033
NAME	GIRIJAN D
PROJECT TITLE	GLOBAL ENERGY CONSUMPTION VISUALIZATION DASHBOARD
DATE OF SUBMISSION	07th APRIL
FACULTY IN-CHARGE	**Mrs. M. Rubina Begam**

Signature of Faculty In-charge

INTRODUCTION

Energy consumption is a key indicator of economic growth and development. With increasing industrialization and population growth, global energy demand has risen significantly. Understanding how energy is consumed across different countries and time periods is crucial for sustainable planning.

This project focuses on analyzing global energy consumption data and presenting insights through interactive visualizations. By using data analytics techniques, the project aims to make complex energy data easy to understand.

PROBLEM STATEMENT

The rapid increase in global energy consumption due to industrialization, urbanization, and population growth has made it difficult to analyze and interpret energy usage patterns effectively using traditional methods. Large volumes of energy data are available from various open sources, but they are often complex, unstructured, and not easily understandable for decision-making.

The challenge is to design and develop a Global Energy Consumption Visualization Dashboard that can transform raw data into meaningful insights through interactive and intuitive visual representations.

The problem involves creating a system that can:

• Collect global energy consumption data from reliable open-source platforms such as Our World in Data
• Process and clean large datasets to ensure accuracy and consistency
• Analyze energy consumption trends across different countries and time periods
• Compare renewable and non-renewable energy usage
• Present the analyzed data through interactive visualizations such as charts, graphs, and dashboards
• Allow users to explore data dynamically using filters (e.g., country, year, energy type)

The system should be user-friendly, efficient, and capable of handling large datasets while providing clear and insightful visual outputs.

This project aims to demonstrate the application of data analytics, data preprocessing, and visualization techniques to solve real-world problems. It also

highlights the importance of visual storytelling in understanding global energy trends and supporting data-driven decision-making.

GOAL:

The main goal of this project is to develop an interactive visualization dashboard that analyzes and presents global energy consumption data in a clear and meaningful way.

The system should be capable of processing large datasets and transforming them into insightful visual representations, enabling users to understand energy usage patterns across different countries and time periods.

Specific goals include:

• To collect and utilize global energy data from reliable sources such as Our World in Data
• To perform data cleaning and preprocessing for accurate analysis
• To analyze trends in energy consumption across countries and years
• To compare renewable and non-renewable energy usage
• To create interactive visualizations such as charts, graphs, and dashboards
• To enable user interaction through filters like country, year, and energy type
• To effectively communicate insights using visual analytics

Ultimately, the project aims to demonstrate how data analytics and visualization techniques can be used to convert complex global data into actionable insights, supporting better understanding and decision-making regarding energy consumption trends.

THEORETICAL CONCEPTS:

1. Data Analytics and Decision-Making

Data analytics involves extracting meaningful insights from raw data to support decision-making. In this project, global energy consumption data is analyzed to identify trends, patterns, and comparisons across countries and time periods. The insights generated help in understanding energy usage and can assist policymakers and researchers in making informed decisions.

---

2. Data Collection and Open Data Sources

Reliable data collection is essential for accurate analysis. This project utilizes datasets from open platforms such as Our World in Data, which provide comprehensive and structured global energy data. These datasets include information on energy consumption, production, and sources across multiple countries and years.

---

3. Data Preprocessing

Raw data often contains missing values, inconsistencies, or irrelevant information. Data preprocessing techniques such as cleaning, filtering, and transformation are applied to ensure the dataset is accurate and suitable for analysis. This step improves the reliability of the results and visualizations.

---

4. Data Visualization

Data visualization is the graphical representation of data to make complex information easy to understand. In this project, charts such as line graphs, bar charts, and pie charts are used to represent energy consumption trends. Visualization helps users quickly identify patterns, comparisons, and anomalies.

---

5. Exploratory Data Analysis (EDA)

EDA is used to summarize and understand the main characteristics of the dataset. It involves analyzing distributions, trends, and relationships within the data. This helps in identifying key insights such as high energy-consuming countries, growth patterns, and shifts towards renewable energy.

---

6. Interactive Dashboards

An interactive dashboard allows users to explore data dynamically. Features such as filters (by country, year, or energy type) enable users to customize the analysis. This improves user engagement and provides a better understanding of the data through real-time interaction.

---

7. Programming and Control Flow

The project utilizes programming concepts such as data structures, loops, and functions to process and analyze data. Libraries in Python are used to handle large datasets efficiently and generate visualizations. Proper control flow ensures smooth execution of data processing and dashboard functionality.

ALGORITHM EXPLANATION WITH EXAMPLE

Algorithm: Global Energy Data Analysis and Visualization

Step-by-Step Algorithm

1. Start
2. Load Dataset
   Import the global energy dataset (CSV file) from sources like Our World in Data.
3. Data Preprocessing
   - Handle missing values
   - Remove irrelevant columns
   - Convert data into proper formats (e.g., year as integer)
4. Filter Data
   - Select required countries
   - Select time period (years)
   - Choose energy type (renewable / non-renewable)
5. Perform Analysis
   - Calculate total energy consumption
   - Compare energy usage across countries
   - Analyze trends over time
6. Generate Visualizations
   - Line chart → Energy consumption over years
   - Bar chart → Country-wise comparison

    - Pie chart → Energy type distribution
7. Display Dashboard
    - Show interactive charts
    - Enable user filters
8. End

Example

Consider a dataset with the following values:

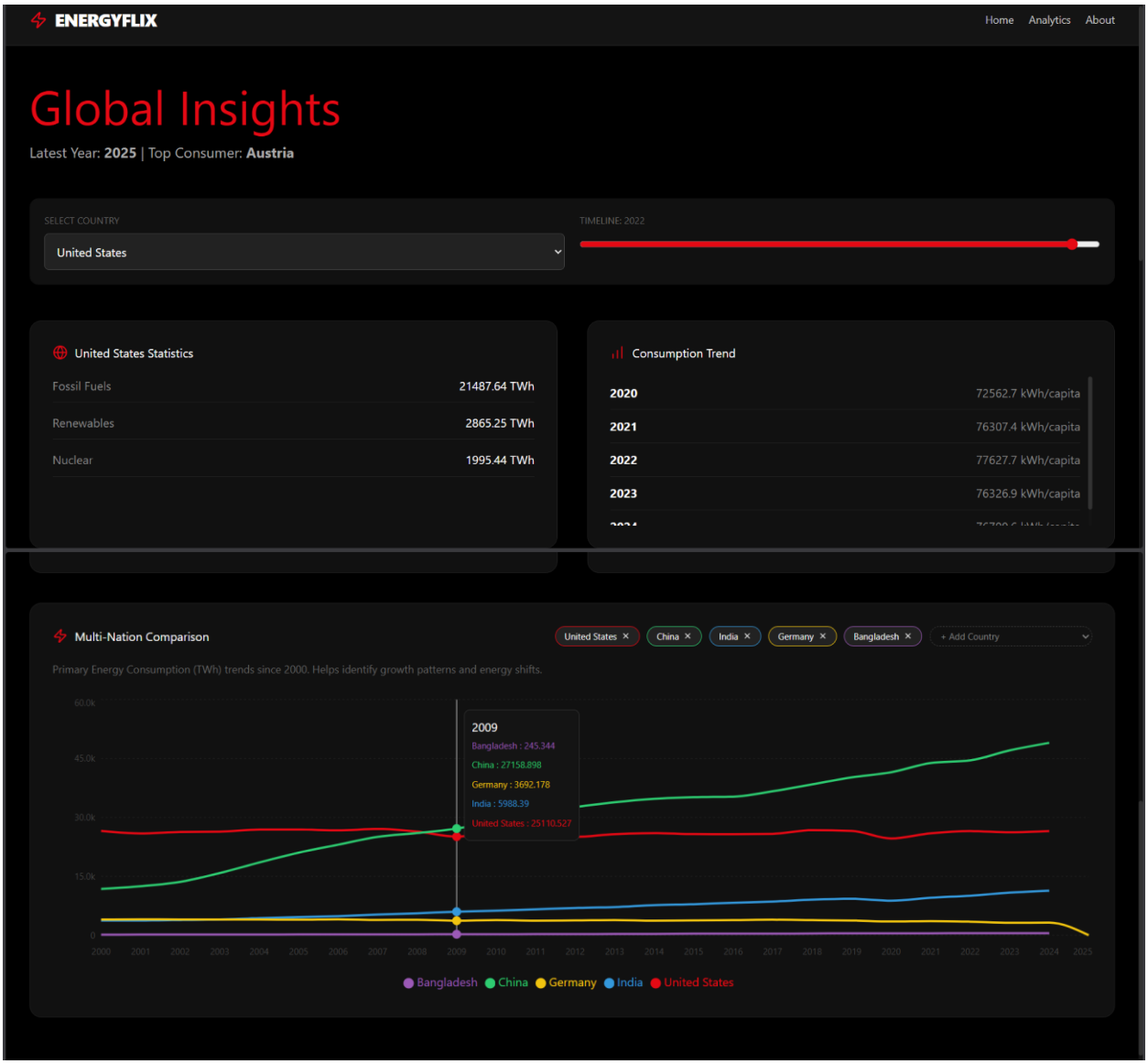
Country	Year	Energy Consumption (TWh)
India	2020	1500
India	2021	1600
USA	2020	4000
USA	2021	4200

IMPLEMENTATION AND CODE

Below is a sample Python implementation used for data analysis and visualization:

```
import pandas as pd
import matplotlib.pyplot as plt
data = pd.read_csv("owid-energy-data.csv")
data = data[["country", "year", "primary_energy_consumption"]]
countries = ["India", "United States"]
filtered_data = data[data["country"].isin(countries)]
filtered_data = filtered_data.dropna()
for country in countries:
    country_data = filtered_data[filtered_data["country"] == country]
    plt.plot(country_data["year"], country_data["primary_energy_consumption"], label=country)
plt.xlabel("Year")
plt.ylabel("Energy Consumption (TWh)")
plt.title("Energy Consumption Comparison")
plt.legend()
plt.show()
```

OUTPUT


ENERGYFLIX
Home
Analytics
About
Global Insights
Latest Year: 2025 | Top Consumer: Austria
SELECT COUNTRY
United States
TIMELINE: 2022
United States Statistics
Fossil Fuels 21487.64 TWh
Renewables 2865.25 TWh
Nuclear 1995.44 TWh
Consumption Trend
2020 72562.7 kWh/capita
2021 76307.4 kWh/capita
2022 77627.7 kWh/capita
2023 76326.9 kWh/capita
Multi-Nation Comparison
United States
China
India
Germany
Bangladesh
+ Add Country
Primary Energy Consumption (TWh) trends since 2000. Helps identify growth patterns and energy shifts.
2009
Bangladesh : 245.344
China : 27158.898
Germany : 3692.178
India : 5988.39
United States : 25110.527
Bangladesh
China
Germany
India
United States

RESULTS AND FUTURE ENHANCEMENT

The Global Energy Consumption Visualization Dashboard was successfully developed and implemented using data analytics and visualization techniques. The system effectively processes large-scale energy datasets obtained from reliable sources such as Our World in Data and transforms them into meaningful visual insights.

The dashboard allows users to explore energy consumption patterns across different countries and time periods through interactive charts and graphs. It provides clear visual comparisons of energy usage, including trends over years and differences between renewable and non-renewable energy sources.

The results demonstrate that:

• The system accurately analyzes and represents global energy consumption data
• Interactive visualizations improve understanding of complex datasets
• Trends such as increasing energy demand and variation across countries are clearly visible
• The dashboard provides an intuitive and user-friendly interface for data exploration

Overall, the project successfully achieved its objective of converting raw energy data into insightful visual representations, enabling better understanding of global energy consumption patterns.

Future Enhancements

Although the current system performs effectively, several improvements can be made to enhance its functionality and impact:

• Integration of real-time energy data using APIs
• Implementation of predictive analytics using machine learning models
• Inclusion of more detailed country-level and region-wise analysis
• Development of a mobile-friendly or responsive dashboard interface
• Addition of advanced visualizations such as heatmaps and geographical maps
• Deployment of the dashboard on cloud platforms for wider accessibility

Git Hub Link of the project and report	https://github.com/Giri6126/Data-Analytics-Project

REFERENCES

- Our World in Data. *Energy Dataset and Documentation.* Available at: https://ourworldindata.org/energy
- World Bank. *World Development Indicators – Energy Data.* Available at: https://data.worldbank.org
- International Energy Agency. *Global Energy Statistics.* Available at: https://www.iea.org
- Python Software Foundation. *Python Documentation.* Available at: https://docs.python.org/3/
- W3Schools. *Python Data Analysis and Visualization Tutorials.* Available at: https://www.w3schools.com/python/
- GeeksforGeeks. *Data Analysis with Python.* Available at: https://www.geeksforgeeks.org
- Real Python. *Data Visualization and Python Guides.* Available at: https://realpython.com/